

ArcShield / CIAER Mathematical Master Set

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Purpose: This document formalizes the mathematical framework underpinning the ArcShield mp4Real™ Reality Codec. It isolates the equations for the codec’s information theory, edge-compute gating, behavioral drift detection, and the physical reward function, ensuring rigorous consistency for Section 6.6 of the whitepaper.

1 Information Theory & The mp4Real™ Codec

The mp4Real™ architecture compresses human operational expertise by recognizing behavioral patterns (causal primitives). The mathematical efficiency of this multiplexing is governed by Information Theory.

1.1 Total Correlation (Multi-Information)

The storage and bandwidth savings achieved by joint-coding the multiplexed tracks (Video, Audio, Biometric, Process Telemetry) under the primitive map π is defined as the Total Correlation conditional on that primitive. For $K > 2$ tracks:

$$TC(T_1, T_2, \dots, T_K \mid \pi) = \sum_{k=1}^K H(T_k \mid \pi) - H(T_1, T_2, \dots, T_K \mid \pi) \quad (1)$$

Where:

- T_k : Individual data tracks locked to a strict Presentation Timestamp (ϵ_{sync}).
- π : The recognized causal primitive (e.g., specific die adjustment).
- $H(\cdot)$: Shannon Entropy.

1.2 Asynchronous Voice Quantization

To strip out non-decision chatter, the backend filters the acoustic track (V_t) by calculating its Mutual Information with the physical action space (A_t):

$$I(V_t; A_t) = H(V_t) - H(V_t \mid A_t) \quad (2)$$

Only vocalizations demonstrating high mutual information with physical interventions are retained in the behavioral codebook.

2 Cyber-Physical LLR Gating (Edge Compute)

The edge application (Pixel device) acts as a computational gatekeeper, utilizing a Neyman-Pearson generalized-likelihood-ratio test to determine which windows of time ($W_{\text{pre}}, W_{\text{post}}$) warrant capture.

2.1 The Log-Likelihood Ratio (LLR) Test

The continuous data buffer is strictly dropped unless the combined log-likelihood ratio (Λ) of an event occurring crosses a predefined threshold (τ):

$$\Lambda \geq \tau \quad (3)$$

Using the generalized-likelihood-ratio approximation for composite hypotheses, this is evaluated as the sum of independent environmental and biometric vectors:

$$\Lambda \approx \Lambda_{\text{env}} + \Lambda_{\text{bio}} \geq \tau \quad (4)$$

2.2 Conditional Independence Assumption

To compute Λ efficiently on the edge without combinatorial explosion, the system assumes that the environmental/telemetry state ($\mathbf{S}_{\text{env}}$) and the operator’s biometric state ($\mathbf{S}_{\text{bio}}$) are conditionally independent given the true event (E) and physical nuisance variables ($\mathbf{Z}$, such as ambient barrel heat):

$$\mathbf{S}_{\text{env}} \perp\!\!\!\perp \mathbf{S}_{\text{bio}} \mid E, \mathbf{Z} \quad (5)$$

3 Reflexivity Trap & Behavioral Drift

To prevent operators from blindly complying with the AI (The Reflexivity Trap), the server continuously audits for behavioral drift to ensure tacit knowledge is still being generated.

3.1 Kullback-Leibler Divergence

The backend calculates the divergence between the baseline expert probability distribution (P) and the incoming operational distribution (Q). If D_{KL} spikes, it triggers a withhold-sampling policy:

$$D_{KL}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \left(\frac{P(x)}{Q(x)} \right) \quad (6)$$

4 Outcome-Grounded Confidence

To further decouple human compliance from the AI’s internal confidence metric, the system’s probabilistic weights are immutably anchored in cyber-physical reality.

4.1 The Confidence Update Equation

The confidence score ($\mathcal{C}$) for an advised action ($\hat{a}_t$) is updated exclusively via an objective physical reward function (R_{phys}) derived from the multiplexed telemetry track ($\mathbf{T}_{t+\Delta t}$). Standard behavioral compliance ($\mathbb{I}(a_t = \hat{a}_t)$) acts only as a gating scalar, not the reward itself:

$$\mathcal{C}_{t+1}(\hat{a}_t \mid s_t) = \mathcal{C}_t(\hat{a}_t \mid s_t) + \alpha [R_{\text{phys}}(\mathbf{T}_{t+\Delta t}) - \mathcal{C}_t(\hat{a}_t \mid s_t)] \cdot \mathbb{I}(a_t = \hat{a}_t) \quad (7)$$

Where:

- α : The learning rate.
- $\mathbf{T}_{t+\Delta t}$: The physical process telemetry at the conclusion of the operational window.
- $\mathbb{I}(a_t = \hat{a}_t)$: The indicator function of human compliance.

4.2 Physical Reward Independence

Crucially, the physical reward evaluated from the state transition is formally independent of the human’s behavioral policy:

$$R_{\text{phys}}(\mathbf{T}_{t+\Delta t}) \perp\!\!\!\perp \mathbb{I}(a_t = \hat{a}_t) \mid s_{t \rightarrow t+\Delta t} \quad (8)$$

This guarantees that confidence only scales when the extrusion line reaches equilibrium, actively penalizing the model if compliance results in a negative physical state transition.